1. CS403 Database Management System

2. https://www.vustd.com/

3. Which is not the function of Database Management System?
    a. A user accessible Catalog
    b. Concurrency Control Services
    c. Failure Response Service
    d. Support for Data Communication
4. The conceptual database design can be transformed into any_____.
    a. Data model
    b. Database design
    c. E-R model
    d. Entity set
5. Which is the second stage in the database development process?
    a. Database design
    b. Preliminary Study
    c. Physical Design
    d. Requirement Analysis
6. Functional dependencies under interference rule reflexivity are called_____dependencies.
    a. Augmentation
    b. Transitivity
    c. Projectivity
    d. Trivial
7. If many instances of the "Book" entity are associated with the many instances of the "Student" entity, then the relationship between them is:
    a. One to one
    b. One to many
    c. Many to many
    d. Many to one
8. Which of the following constraints enforces entity integrity?
    a. PRIMARY KEY
    b. FOREIGN KEY
    c. CHECK
    d. NOT NULL
9. Unary operations involves
    a. Only one relation
    b. Only two relations
    c. More than two relations
    d. Not more than five relations
10. A_____entity has a primary key that is partially or totally derived from the parent entity in the relationship.
    a. Strong
    b. Weak

c. Business
d. Relationship

11. In_____appropriate data model is chosen.
    a. ERD
    b. Conceptual Database Design
    c. Logical Database Design
    d. DFA
12. Which of the following is drawback of standardization?
    a. Difficult to understand
    b. Uniformity
    c. Lack of uniqueness
    d. Easy to develop applications
13. Recursive relationship is also called_____.
    a. Unary
    b. Ternary
    c. Primary
    d. Secondary
14. The foreign key attribute, which is present as a primary key in another relation is called as _____of the foreign key attribute.
    a. Binary relation
    b. Composite relation
    c. Home relation
    d. Recursive relation
15. DFD does not provide us a way to represent _____
    a. Entities
    b. Decisions
    c. Date stores
    d. Processes
16. A_____is used to maintain a connection between the users of the database system.
    a. Virtual server
    b. File server
    c. Client server
    d. FTP server
17. Which of the following is an advantage of using the Traditional File Organization?
    a. Time consuming
    b. Data security
    c. Simplicity
    d. Efficiency
18. Which of the following is the input to Analysis Phase?
    a. DFD
    b. Logical database design
    c. Physical database design
    d. Data model
19. Name and_____are part of the definitions of an attribute.

a. **Domain**
b. Symbol
c. Property
d. Entity instance

20. Which of the following most certainly implies the need for an entire table to implement?
    a. **A ternary relationship**
    b. A recursive relationship
    c. An identify relationship
    d. A binary relationship
21. Core of the Database Architecture is _____
    a. **Internal level**
    b. External level
    c. Conceptual level
    d. ANSI SPARK
22. _____is same as equi-join with a slight difference.
    a. **Natural join**
    b. Semi join
    c. Outer join
    d. Theta join
23. Conceptual database design is implemented using a_____.
    a. Hierarchical data model
    b. **Semantic data model**
    c. Relational data model
    d. Network data model
24. Which constraint is generally used for efficiency purposes in the data entry process?
    a. Null value
    b. Integer value
    c. **Default value**
    d. String value
25. A 3 NF relation is converted to BCNF by
    a. Removing composite keys
    b. Removing multivalued dependencies
    c. **Dependent attributes of overlapping composite keys are put in a separate relation**
    d. Dependent non-key attributes are put in a separate table
26. Each _____of a table contains atomic/single value.
    a. **Cell**
    b. Row
    c. Column
    d. Record
27. A relation is said to be in BCNF when
    a. It has overlapping composite keys
    b. It has no composite keys
    c. It has no multivalued dependencies
    d. **It has no overlapping composite keys which have related attributes**

28. Which one of the following is not a part of three-level architecture?
    a. External level
    b. Conceptual level
    c. Internal level
    d. Physical level
29. Which of the following is INCORRECT about naming entities?
    a. Singular nouns are used
    b. Write in capitals
    c. Organization specific names can be used
    d. Always use abbreviations as they are convenient
30. An instance is
    a. A particular occurance of an entity
    b. A special type of relation
    c. An attribute of an entity
    d. Any particular entity
31. _____is an example of metadata
    a. Ali
    b. Address
    c. Person
    d. Student
32. Level of data at which entities or objects exist in reality is called
    a. Real world data
    b. Schema for real world data
    c. Meta data
    d. Data about data
33. There are_____basic properties of the database relations.
    a. Four
    b. Three
    c. Six
    d. Two
34. How many types of dependencies are?
    a. One
    b. Two
    c. Three
    d. Four
35. In_____join, only selected rows of a relation are made cross product with second relation.
    a. Normal
    b. Theta
    c. Outer
    d. Semi
36. Which of the following is not a key?
    a. Super key
    b. Alternate key
    c. Integrity constraint key

d. Primary key

37. A collection of concepts that can be used to describe the structure of a database
    a. Database
    b. DBMS
    c. Data model
    d. Data
38. Any combination of attributes with super key is ____
    a. Primary key
    b. Foreign key
    c. Super key
    d. Candidate key
39. Binary relationships are those, which are established between
    a. One entity type
    b. Two entity type
    c. Three entity type
    d. Unlimited entities
40. _____ are called participants, when enrolled in a relationship.
    a. Attributes
    b. Relations
    c. Entities
    d. Keys
41. Incase of Context-level Diagram, the system is represented by
    a. One process atleast
    b. Two processes atleast
    c. One process only
    d. Any number of processes
42. Collection of raw facts is called_____.
    a. Information
    b. Object
    c. Output
    d. Data
43. A software package designed to store and manage database
    a. Database
    b. DBMS
    c. Data model
    d. Data
44. In_____join common attributes appear twice in the output with selected rows.
    a. Normal join
    b. Semi join
    c. Natural join
    d. Equi join
45. Within a table, each primary key value_____.
    a. Is a minimal super key
    b. Is always the first field in each table

c. Must be numeric
d. Must be unique

46. _____is not an operation of relational algebra.
a. Addition
b. Union
c. Cartesian product
d. Set difference

47. Candidate keys which are not selected as primary key are called _____
a. Secondary keys
b. Foreign keys
c. Alternate keys
d. Composite keys

48. Major benefits of data sharing is:
a. Data of same applications placed at same place.
b. Data of same applications placed at different places
c. Data of different application places at same place
d. Data of different applications stored at placed place

49. Functional dependency between two attributes or sets of attributes i.e. Q and P is represented as_____.
a. $Q \rightarrow P$
b. $Q \ominus P$
c. $Q * P$
d. $Q + P$

50. Which of the following is not the component of Relational Data Model?
a. Integrity Constraint
b. Manipulation Language
c. Relation / Table
d. Domain Constraint

51. The attribute that is calculated from other attribute is called_____.
a. Derived attribute
b. Multi-valued attribute
c. Composite attribute
d. Simple attribute

52. Consider the two attributes "CNIC_NO" and "Registration_NO". Both the attributes can be used for the unique identification. Now if the user selects the "CNIC_NO" as primary key, then "Registration_NO" is a____key.
a. Foreign
b. Secondary
c. Primary
d. Alternative

53. Which of the following constraint is included in the definition of the table?
a. Default value
b. Foreign key
c. Null

d. Domain

54. The database deals with large amount of data. This property is called
    a. Database design
    b. Scalability
    c. Concurrency
    d. Robustness
55. Which one of the following is NOT a characteristic of meta data?
    a. Data about data
    b. Describes a data dictionary
    c. Self-describing
    d. Includes user data
56. Theta join is denoted mathematically as_____.
    a. R X S
    b. R X Θ
    c. R X Θ S
    d. R X Θ RS
57. In cross reference matrix transitions are specified on_____and database objects are specified on_____.
    a. Y axis, X axis
    b. X axis, Y axis
    c. X axis, X axis
    d. Y axis, Y axis
58. Each component of DBMS performs_____functions.
    a. Similar
    b. Different
    c. Identical
    d. Computational
59. Which of the following data model is not record based?
    a. Hierarchical Data Model
    b. Network Data model
    c. Semantic Data model
    d. Relational Data model
60. The basic purpose of diamond in ER diagram is to show the_____.
    a. Attributes
    b. Relationships
    c. Composite attribute
    d. Entity set
61. Which of the following tool is used to design conceptual database design?
    a. SQL
    b. MS Access
    c. Oracle
    d. This model is independent of any tool
62. In "one to many" cardinality one instance of a relation is mapped with_____.
    a. Many instances of second entity type

b. One instance of second entity type
c. Primary key of second relation
d. Foreign key of second relation

63. Cross reference matrix is developed at the _____ stage of the database.
    a. Analysis
    b. Designing
    c. Implementation
    d. Application
64. The population of the data of the organization for which the database is created is called the _____ of the database.
    a. Extension
    b. Intension
    c. Model
    d. Schema
65. User rights information is stored in
    a. Physical database
    b. Catalog
    c. Logical database
    d. Buffer
66. Logical data independency provides independency to change _____ model.
    a. External
    b. Logical
    c. Conceptual
    d. Internal
67. In natural join the common attribute in the output table appears _____.
    a. Twice
    b. Three times
    c. Four times
    d. Once
68. Which levels are mostly used for detailed DFD?
    a. Level-0, Level-1
    b. Level-1, Level-2
    c. Level-2, Level-3
    d. Level-3, Level-4
69. By default, a non-key attribute in a relation can have _____ value.
    a. Binary
    b. Null
    c. Character
    d. Integer
70. The three levels architecture is useful for _____.
    a. Arranging the data in an organized manner
    b. Hiding the details of internal systems
    c. Breaking the data access restriction
    d. Storing definitions of the structures

71. With the help of_____technique data values with the smaller sized codes can further reduce the space needed by the data for storage in the database.
    - **a. Compression**
    - b. Expansions
    - c. Negation
    - d. Extension
72. If an entity is linked with itself, than it is called_____relationship.
    - a. Binary
    - b. Ternary
    - c. Nary
    - **d. Recursive**
73. Duplication of data in controlled redundancy is controlled and_____.
    - **a. Deliberate**
    - b. Necessary
    - c. Precise
    - d. Unintentional
74. Teleprocessing is a sub-type of_____.
    - a. Single user database
    - **b. Multi-user database**
    - c. File server
    - d. Client server
75. In an ERD, the focus is on the _____and the relationships between them.
    - a. Attributes
    - **b. Entities**
    - c. Keys
    - d. Indexes
76. The most widely used conceptual model is the_____model.
    - a. Implementation
    - **b. ER**
    - c. Object oriented
    - d. Internal
77. Student and book are two entities, what is the cardinality between the two?
    - a. Many to one
    - b. One to many
    - **c. Many to many**
    - d. One to one
78. RDM is a simple there is just one structure and that is_____.
    - **a. A relation or a table**
    - b. E-R model
    - c. Attributes
    - d. Data model
79. A description on a particular collection of data using the given data model.
    - ~~a. Database~~
    - b. Relation
    - c. Schema
    - d.

None

80. Consider thy key "CNIC, Name" and choose the best option that describes the key most effectively.
    a. The given key is a super key and alternate key.
    b. The given key is not a super key but a candidate key.
    c. The given key is a super key and candidate key as well.
    d. The given key is a super key but not a candidate key.
81. Normalization is a process of restricting a relation to
    a. Minimize duplication of data in a database
    b. Maximize duplication of data to ensure reliability
    c. Make if of uniform size
    d. Allow addition of data
82. Unary relationship involves a single entity it is also called _____.
    a. One to one relationship
    b. One to many relationship
    c. Many to many relationship
    d. Recursive relationship
83. Which of the following is a part of a DFD but not included in the ERD?
    a. The attribute having a single instance
    b. The external entity having a single instance
    c. The relationship having a single instance
    d. The role having a single instance
84. The database management system (DBMS) is used to
    a. Store the data
    b. Access the data
    c. Manage its users
    d. All of above
85. In cross reference matrix attributes are mention on_____and reports are mention on _____.
    d. Y axis, X axis
    e. Y axis, Z axis
    f. X axis, Y axis
    g. X axis, Z axis
86. A table can be logically connected to another table by defining a _____.
    a. Hyper link
    b. Common attribute
    c. Primary key
    d. Super key
87.
88. In the relation model, the number of attributes and number of tuples in a relation are termed as _____ and _____respectively.
89.
90. Answer: degree, cardinality
91.
92. The function that an entity plays in a relationship is called that entity's _____.
93.

94. Answer: Role

95.

96. A database system is designed to meet the information needs of _____ in an organization.

97.

98. Answer: Multiple users

99.

100. Which of the following is the example of a strong entity?

101.

102. Answer:

103.

104. Ahmad account of ABC bank has no existence if the ABC bank doesn't exist anymore

105. (Not sure)

106.

107. Which of the following is a completeness constraint?

108.

109. Answer: Partial Completeness Constraint

110.

111. In an employee entity which of the following attribute will work as a primary key?

112.

113. Answer: Employee id

114.

115. _____ is the first computer-based approach for handing the commercial or business applications

116.

117. Answer: Database Systems

118.

119. Which of the following is NOT the classification of entity type?

120.

121. Answer: Dependent entity type

122.

123. Which of the following is NOT the type of key?

124.

125. Answer: Derived key

126. Data Redundancy means

a. Security of data
b. **Duplication of data**
c. Management of data

d. Recovery of data

127. Which of the following is a type of dependency?
   a. DBMS
   b. ERD
   c. DFD
   d. Referential
128. In your opinion, why relational database is widely acceptable?
   a. Due to its complexity
   b. Due to its approach
   c. Due to its dependencies
   d. Due to its strength
129. _____is the first comprehensive complete database design.
   a. Logical database design
   b. Relational database design
   c. Conceptual database design
   d. Hierarchical database design
130. How data security can be assured in the database?
   a. By hiding data from all
   b. Through concurrent access of users
   c. By asking users not to share things
   d. Implementing encryption algorithms
131. The attributes whose value is not stored in a database is known as _____attribute.
   a. Stored
   b. Single value
   c. Multi value
   d. Derived
132. Logical database design, like conceptual database design is our _____
   a. Data model
   b. Weak entity set
   c. Strong entity set
   d. Database design
133. If a single instance of "student" is associated with the many instances of the "subjects" entity, then the relationship between them is:
   a. One to one
   b. One to many
   c. Many to many
   d. Many to one
134. The conceptual database design is drawn through_____.
   a. Data model
   b. ER model
   c. Database design
   d. Entity set
135. A relational database is
   a. One that consist of two or more tables
   b. A database that is able to process tables, queries, forms, reports and macros

c. One that consists of two or more tables that are joined in some way

d. The same as a flat file database

136. Database is a computerized representation of any organization's flow of_____and storage of _____.

a. Data, information

b. Details, data

c. Data, details

d. Information, data

137. In which stage of the database development process logical design is created?

a. Requirement analysis

b. Database design

c. Physical design

d. Implementation

138. Which of the following is NOT an entity?

a. Employee

b. Hobby

c. Student

d. Playground

139. After moving the determined columns from the original table to the new table while converting to 2NF, the determinate becomes the____of the new table.

a. Primary key

b. Foreign key

c. Secondary key

d. Composite key

140. The select operation works horizontally on the table, on the other hand the _____ operates on a single table vertically.

a. Binary operator

b. Project operator

c. Join operator

d. Select operator

141. Free standing data dictionary is created by

a. DFD

b. Database

c. CASE tools

d. Cross reference matrix

142. A patient can be an outdoor or indoor. Analyze this statement and tell what type of constraint will be applied on patient?

a) Total completeness

b) Partial completeness

c) Total disjoint

d) Partial disjoint

143. If A → B and A → C, then A → BC. The inference Rule applies is:

a) Union

b) Decomposition

c) Augmentation
d) Reflexivity

144. Which of the following is the correct abbreviation of ERD?
a) Entity relationship diagram
b) Entity relation diagram
c) Entity reality diagram
d) Entity real distance

145. Database relation is also represented in a two dimensional structure called _____
a) Row
b) Table
c) Entity
d) Column

146. Data about data is metadata
a) True
b) False

147. Which of the following describes the job of a database administrator?
a) Development, implementation operation of the physical database
b) Creation of the system catalog
c) Monitoring and controlling database security and authorization: setting up controls to ensure the quality and integrity of data
d) All of the above

148. Each table must have a _____ key.
a) Primary
b) Secondary
c) Logical
d) Foreign

149. A relational database system is based on the concept(s) of:
a) A network of data records
b) Nested tables
c) Tables, row and columns
d) A tree-like structure of data

150. $A \rightarrow BC$ and $A \rightarrow B$, then $A \rightarrow C$ follow _____ inference rule.
a) Augmentation
b) Transitivity
c) Projectivity
d) Additivity

151. Suppose, we have two entities "shape" and "triangle", then which of the following is a correct statement?
a) Shape is a super type entity and triangle is a sub type entity
b) Shape is a sub type entity and triangle is a super type entity
c) Shape and triangle both are super type entities
d) Shape and triangle both are sub type entities

152. The entity that is NOT dependent on any other entity for its existence is known as _____ entity.

a) Weak
b) Strong
c) Derived
d) Dependent

153. In cross reference matrix attributes are specified on _____
a) X axis
b) Y axis
c) Z axis
d) None of the above

154. Which of the following clarify the semantics of a relationship?
a) Entity
b) Roles
c) Relationship set
d) Participants

155. How many types of cardinality is specified about relationship?
a) Two
b) One
c) Three
d) Four

156. Which of the following is NOT a component of a DFD?
a) Dataflow
b) Datastore
c) External entities
d) Relationship between external entities

157. Which of the following provides us the facility to add properties of one entity to another entity automatically?
a) Abstraction
b) Encapsulation
c) Inheritance
d) Dependencies

158. If there is more than one relationship between two entities and we mentioned role on the relationship link, then it is called:
a) Unary relationship
b) Binary relationships
c) Ternary relationships
d) Multiple relationships

159. Which one of the following is not an advantage of DBMS
a) Data consistency
b) Recovery procedure is difficult
c) Faster development of new applications
d) Better backup and recovery procedures

160. In many to many relationship a third table is created for the relationship, which is also called as_____
a) Binary relation

b) Associative entity type

c) Unary relationship

d) Composite attributes

161. When one entity instance of another entity for its existence, then it is called _____

a) Existence dependency

b) Identifier dependency

c) Referential dependency

d) Acceptance dependency

162. Relational data model is widely accepted due to its mathematically proven foundation and its _

a) Complexity

b) Simplicity

c) Hierarchical structure

d) Traditional structure

163. If an entity "motor cycle" is associated with at most one "driver", then the relationship between these two entities is:

a) One to one

b) One to many

c) Many to many

d) Many to one

164. Database application is a program which is used for performing certain operations such as

a) Insertion of data

b) Extraction of data

c) Updating the data

d) All of above

165. Creating the relationships between new tables and their predecessors through the use of foreign keys is general requirements of _____

a) First normal form

b) Second normal form

c) Third normal form

d) Fourth normal form

166. In_____the domains of attributes of a relation are atomic, that is they consist of single units that cannot be broken down further.

a) First normal form

b) Second normal form

c) Third normal form

d) Augmentation

167. Atomicity is a feature of_____.

a) 1NF

b) 2NF

c) 3NF

d) BCNF

168. The_____can see entire information structure of the database.

a) **DBA**
b) DBTG
c) Database user
d) External viewer

169. The attribute whose value is not stored in a database is known as _____ attribute.
a) Stored
b) Single value
c) Multi value
d) **Derived**

170. _____ are the structure defined for placing data in the attributes while designing physical design of database.
a) Objects
b) Variables
c) Attributes
d) **Data types**

171. Suppose a relation has five columns and ten rows, select the correct option to identify cardinality of relation.
a) Five
b) **Ten**
c) Fifteen
d) Twenty

172. Which feature of database provides conversion from inconsistent state of DB to a consistent state ensuring minimum data loss?
a) User accessible catalog
b) Data processing
c) **Recovery service**
d) Authorization service

Consider an employee is allotted an organization vehicle, which might soley be driven by that employee. What kind of relationship exists between employee and company?

Answer: Many to one

Which of the following is a feature of CHECK constant?

Answer: used to enforce referential integrity

Which of the following semantic model is used for the graphical representation to the conceptual database design?

Answer: Entity Relationship (ER)

Suppose we have two entities department and University. Which type of dependency exists between these two entities?

Answer: Referential (Not sure)

_____ is the next stage of conceptual database design.

Answer: logical database design

The _____ model is independent of any tool or data model.

Answer: Entity Relationship (ER) (page 122)

While converting to 2NF all the columns are deleted from the original table except for the determinate which will serve as a _____ key.

Answer: Primary key

In relational algebra, which of the following operator is used to find tuples that are present in one relation but are not in another?

Answer: Union

Subject may be the prerequisite of another subject, is the example of _____ relationship.

Answer: Unary (google)

173. The maximum PL/SQL size of data type "VARCHAR2" in DBMS is _____bytes
   a) 42767
   b) 52767
   c) 32767
   d) 28767

174. _____is state of database where a "course" cannot be inserted in the table, because this course has not been registered to any "student".
   a) Insertion anomaly
   b) Deletion anomaly
   c) Updation anomaly
   d) Redundancy

175. The maximum PL/SQL size of data type "BLOG" in DBMS is_____
   a) 4 gigabytes
   b) 8 gigabytes
   c) 10 gigabytes
   d) 12 gigabytes

176. _____states that in a relation no attributes of a primary key (PK) can have a null value.
   a) Referential integrity constraint
   b) Null constraint
   c) Domain constraint
   d) Entity integrity constraint

177. A database state where deletion of the information about the student record deletes the course information as well is called_____.
   a) Insertion anomaly
   b) Updation anomaly
   c) Redundancy
   d) Deletion anomaly

178. An entity type is
   a) Defined when the database is actually constructed
   b) A specific type such as an integer, text, date, logical etc.
   c) A coherent set of similar objects that we want to store data on (e.g. STUDENTS, COURSE, CAR)
   d) Defined by the database designer

179. "select" and "project" are the examples of _____
   a) Unary operation
   b) Binary operation
   c) Ternary operation
   d) Nary operation

180. External scheme evolves as user needs are _____over the time.
   a) Ends
   b) Modified
   c) Switch
   d) Stable

181. An attribute that is the collection of some other attribute(s) is known as _____ attributes.
   a) Multi-valued
   b) Single-valued
   c) No-valued
   d) Composite

182. Which of the following is correct regarding Dataflow diagram?
   a) Single DFD is required to represent a system
   b) The dataflow must be bidirectional
   c) Created at increasing levels of detail
   d) Used to represent the relationships among the external entities

183. _____are those attributes that are a combination of two or more than two attributes.
   a) Default value
   b) Null value
   c) Primary key

d) Composite attributes

184. Controlling redundancy in a database management system DOES NOT help to
a) avoid duplication
b) avoid unnecessary wastage of storage space
c) avoid unauthorized access to data
d) avoid inconsistency among data

185. Which of the following concepts is applicable with respect to 3NF?
a) Full functional dependency
b) Any kind of dependency
c) Transitive dependency
d) Partial functional dependency

186. Identify the operation which is NOT one of the parts of the five basic set operations in relational algebra?
a) Join
b) Union
c) Cartesian Product
d) Set Difference

187. Making a change to the conceptual schema of a database but not affecting the existing external schemas is an example of
a) Physical data independence.
b) Concurrency control.
c) Logical data independence.
d) Functional dependency

188. For a third normal form we concentrate on relations with one_____key, and we eliminate transitive dependencies
a) Candidate
b) Composite
c) Super key
d) Primary key

189. The entity relationship diagram is used to graphically represent the_____database model.
a) Condensed
b) Physical
c) Logical
d) Conceptual

190. The_____determines that the link between two entities is optional or mandatory.
a) Maximum cardinality
b) Minimum cardinality
c) Maximum degree
d) Minimum degree

191. Select the most appropriate statement. Which of the following data model is used to design logical database design?
a) Relational
b) Hierarchal

c) Object oriented
d) Any model can be used

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_____ are used by database administrator in situations where we have large number of records and wastage of small amount of space in each record can lead to
loss of huge amount of data storage space.

Answer: Codes

An attribute in a relation is foreign key If the _____ key from one relation is used as an attribute in that relation.

Answer: primary

in data flow diagram which of the following operator is used, when data flows from a source process to one of the mentioned sinks?

Answer: Ring Sum Operator

The _____ rule specifies that an entity can be a member of only one subtype at a time.

Answer: disjoint page no 108

In relational algebra, which of the following Greek is used to represent the project operator?

Answer: ∏ (PI) (page no 150)

In three level architecture of the database, which views is called community

views of the database?

Answer: Conceptual view (page no 37)

Which of the following normal forms deals multivalued dependency?

Answer: 4NF

While mapping _____ relationship, primary key of the entity type against one side of relationship will be included in the entity type on the many side of the relationship as foreign key.

Answer: one to many

In data flow diagram _____ symbol is used to represents data flows.

Answer: arrow (google)

In the below entity type which attribute can work as an alternate key?

Employee (Reg# Primary key,Name,CNIC,designation,address,phone)

Answer: CNIC

From the following who is responsible for proper working of the database and DBMS?

Answer: Database administrator

Suppose that "Program" is an entity which represents different courses are being offered by an institute, like MCS. BCS etc. Following are the major attributes of this entity.
Identify the attribute which can be used as primary key.

Answer: program_ Code

The following is an of _____ inference rule.
If stId --> prName and prName --> credits then stId --> credits

Answer: Transitivity

Which of the following is the example of a weak entity?

Answer: The tire can exist without being attached to the car

In Relational Data Model, the _____ of a relation is the number of columns in that relation

Answer: Degree

Which of the following normal form deals with possible loss less decompositions?

Answer: 5NF

Suppose we have a list of bank customers which availed loan facility from different branches of a bank. If we want to get a list of all customers and the amount of the loans excluding branch name. Which of the following operation allows us to generate this relation?

Answer: Union

Which of the following provides us data independency?

Answer: 3-Level Architecture

Cross reference matrix is a tool available in the data dictionary, which helps us in finding ____.

Answer: Entities of the database and their associations

Conceptual database design, expressed in E-R data model can be implemented using any _____.

Answer: Database Management System (DBMS)

Which of the following might be represented with a single-valued attribute?

Answer: Computer's processor speed

Which of the following is a Unary operation?

Answer: Project

Which of the following data model has the following components?
1. Construct (Tables)
2. Manipulation language (SQL)
3. integrity constraints?

Answer: Relational

Which of the following Greek is used to represent the select operation?

Answer: σ (page 149)

Data about data is called metadata.

Answer: True (page no 10)

Normalization of database is used to _____.

Answer: Eliminate redundancy

In student entity which of the following is the best example of multivalued attribute?

Answer: Email address

Theta join is denoted mathematically as

Answer: R X theta S

In which normal form, we convert composite attributes into individual attributes?

Answer: 1NF

Which of the following tool is used to graphically represent the components and design of the system?

Answer: DFD

Data type "BLOB" in database management system stands for _____.

Answer: Binary Large Object (page no 185)

In a three level DBMS architecture, the _____ level interacts directly with the

users.

Answer: External

In the Database System _____ provides many features to ensure the data integrity.

Answer: Database Management System (DBMS)

The process of defining one or more subtypes of a super and forming relationships is known as

Answer: Generalization

External scheme evolves, as user needs are _____ over the time.

Answer: Switch

_____ of the Database is most difficult to change when a database is created.

Answer: Extension

In Database development process, which phase involves designing of any new application for the enhancement of the system?

Answer: Maintenance phase

An attribute of an entity is associated with another attribute of the same entity is called:

Answer: Recursive relationship

Relational Data model is created in the _____ phase.

Answer: Analysis

Database system saves lots of efforts and finances, this advantage provides _____.

Answer: Time saving structure

Data inconsistency leads to a number of a problems such as _____.

Answer: loss of information and incorrect results

A row of a table represents a _____, whereas the column of the table represents the _____.

Answer: Record, attribute

Every primary key (PK) will be a candidate key and every candidate key will be a _____.

Answer: Super key

If we want to include attributes from two different relations, which of the following operations will be used?

Answer: Union

When have two entities "Person" and "Doctor. Which of all is best suitable type of relationship between the mentioned entities?

Answer: Sub type (Not sure)

The relationship between department to employee is _____.

Answer: one to many